# Phase 2 Modeling Summary

Phase 2: Full-dataset modeling summary

Dataset: all 2,880 rows

Validation: 70/30 stratified holdout + repeated grouped CV on training split

Feature sets compared: embeddings only vs embeddings plus region and Dynamic World context

Current implementation differs from the original proposal: it uses the full dataset and a LightGBM ablation rather than the earlier 720-row subset plus ridge baseline plan.

Best held-out feature set by target:

- S1_VH: Embeddings + region + Dynamic World with R2=0.943, RMSE=1.175, MAE=0.858, r=0.971

- S1_VV: Embeddings + region + Dynamic World with R2=0.902, RMSE=1.260, MAE=0.824, r=0.951

- S1_VV_div_VH: Embeddings only with R2=0.707, RMSE=1.139, MAE=0.796, r=0.841

Interpretation:

- S1_VV is best with Embeddings + region + Dynamic World.

- S1_VH is best with Embeddings + region + Dynamic World.

- VV/VH ratio is best with Embeddings only.

- Mean grouped-CV R2 across the selected models ranges from 0.729 to 0.939.

- Important caveat: the outer holdout is stratified random within sampled regions, not spatially disjoint. Treat these as within-region generalization estimates rather than strict geographic transfer performance.

Derived linear VV/VH ratio baselines:

# Phase 2 Modeling Summary (cont.)

- The stored S1_VV_div_VH target is a dB-space difference, so the new baseline reconstructs the physical VV/VH ratio from VV and VH before fitting/evaluation.

- Best overall linear-ratio baseline: LightGBM direct ratio (Embeddings only) with R2=0.584, RMSE=3.341, MAE=1.502, r=0.765.

- Ridge direct ratio best result: Ridge direct ratio (Embeddings + region + Dynamic World) (R2=0.534).

- Ridge log-ratio best result: Ridge log-ratio (Embeddings + region + Dynamic World) (R2=0.546).

- LightGBM direct ratio best result: LightGBM direct ratio (Embeddings only) (R2=0.584).

# Held-out Performance by Feature Set

| target | feature_set | cv_r2_mean | cv_rmse_mean | stability_r2_mean | stability_rmse_mean | r2 | rmse | mae | pearson_r |
|---|---|---|---|---|---|---|---|---|---|
| S1_VH | Embeddings + region + Dynamic World | 0.939 | 1.220 | 0.938 | 1.229 | 0.943 | 1.175 | 0.858 | 0.971 |
| S1_VH | Embeddings only | 0.940 | 1.211 | 0.939 | 1.223 | 0.941 | 1.194 | 0.861 | 0.970 |
| S1_VV | Embeddings + region + Dynamic World | 0.908 | 1.162 | 0.907 | 1.163 | 0.902 | 1.260 | 0.824 | 0.951 |
| S1_VV | Embeddings only | 0.910 | 1.143 | 0.910 | 1.145 | 0.900 | 1.272 | 0.834 | 0.949 |
| S1_VV_div_VH | Embeddings only | 0.731 | 1.057 | 0.729 | 1.060 | 0.707 | 1.139 | 0.796 | 0.841 |
| S1_VV_div_VH | Embeddings + region + Dynamic World | 0.732 | 1.056 | 0.730 | 1.059 | 0.699 | 1.154 | 0.798 | 0.836 |

The holdout test set is shared across the two feature-set ablations. It is a stratified random holdout within sampled regions rather than a spatially disjoint holdout, so these estimates may be optimistic relative to stricter geographic transfer tests.

# Polarization Difference Experiments

Polarization Difference Experiments

This addendum compares direct and structural approaches for the stored polarization-difference target S1_VV_div_VH = S1_VV - S1_VH.

Best Ridge direct model: Ridge (VV-VH) [Embeddings + region + Dynamic World] with R2=0.708, RMSE=1.137, MAE=0.801.

Best LightGBM direct model: LightGBM (VV-VH) [Embeddings only] with R2=0.707, RMSE=1.139, MAE=0.796.

Best structural baseline: Structural baseline (VV_hat - VH_hat) [Embeddings + region + Dynamic World] with R2=0.708, RMSE=1.137, MAE=0.801.

Interpretation:

- Direct VV-VH prediction tests whether the embedding captures polarization contrast as its own target.

- The structural baseline tests whether separate VV and VH Ridge models preserve the difference after subtraction.

# Polarization Difference Comparison

| feature_set | model_label | target | training_target | r2 | rmse | mae | pearson_r |
|---|---|---|---|---|---|---|---|
| Embeddings only | Ridge (VH) [Embeddings only] | S1_VH | S1_VH | 0.929 | 1.317 | 0.945 | 0.964 |
| Embeddings only | Ridge (VV) [Embeddings only] | S1_VV | S1_VV | 0.868 | 1.456 | 0.938 | 0.933 |
| Embeddings only | Ridge (VV-VH) [Embeddings only] | S1_VV_div_VH | S1_VV_div_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings only | Structural baseline (VV_hat - VH_hat) [Embeddings only] | S1_VV_div_VH | S1_VV+S1_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings only | LightGBM (VV-VH) [Embeddings only] | S1_VV_div_VH | S1_VV_div_VH | 0.707 | 1.139 | 0.796 | 0.841 |
| Embeddings + region + Dynamic World | Ridge (VH) [Embeddings + region + Dynamic World] | S1_VH | S1_VH | 0.929 | 1.315 | 0.944 | 0.964 |
| Embeddings + region + Dynamic World | Ridge (VV) [Embeddings + region + Dynamic World] | S1_VV | S1_VV | 0.870 | 1.445 | 0.934 | 0.934 |
| Embeddings + region + Dynamic World | Ridge (VV-VH) [Embeddings + region + Dynamic World] | S1_VV_div_VH | S1_VV_div_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings + region + Dynamic World | Structural baseline (VV_hat - VH_hat) [Embeddings + region + Dynamic World] | S1_VV_div_VH | S1_VV+S1_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings + region + Dynamic World | LightGBM (VV-VH) [Embeddings + region + Dynamic World] | S1_VV_div_VH | S1_VV_div_VH | 0.699 | 1.154 | 0.798 | 0.836 |

The structural baseline subtracts Ridge VV and Ridge VH predictions and evaluates the result against the stored dB-space polarization difference target.

# VV/VH Ratio Baselines

Phase 2 ratio-baseline addendum

These rows compare direct-ratio, log-ratio, and structural baselines on the derived linear VV/VH target.

The target is computed from the existing VV and VH columns and evaluated on the linear ratio scale.

Best overall baseline: LightGBM direct ratio (Embeddings only) with R2=0.584, RMSE=3.341, MAE=1.502, r=0.765.

Model families in this addendum:

- Ridge direct ratio: fit Ridge directly on VV/VH.

- Ridge log-ratio: fit Ridge on log(VV/VH), then exponentiate predictions.

- LightGBM direct ratio: fit LightGBM directly on VV/VH.

- Ridge structural ratio: predict VV and VH separately with Ridge, then reconstruct VV/VH.

# Derived Linear VV/VH Baselines

| feature_set | model_label | training_target | r2 | rmse | mae | pearson_r |
|---|---|---|---|---|---|---|
| Embeddings only | LightGBM direct ratio (Embeddings only) | S1_VV_div_VH_linear | 0.584 | 3.341 | 1.502 | 0.765 |
| Embeddings only | Ridge log-ratio (Embeddings only) | S1_log_VV_div_VH | 0.539 | 3.516 | 1.454 | 0.755 |
| Embeddings only | Ridge structural ratio (Embeddings only) | S1_VV+S1_VH | 0.539 | 3.516 | 1.454 | 0.755 |
| Embeddings only | Ridge direct ratio (Embeddings only) | S1_VV_div_VH_linear | 0.527 | 3.561 | 1.784 | 0.726 |
| Embeddings + region + Dynamic World | LightGBM direct ratio (Embeddings + region + Dynamic World) | S1_VV_div_VH_linear | 0.569 | 3.401 | 1.492 | 0.755 |
| Embeddings + region + Dynamic World | Ridge log-ratio (Embeddings + region + Dynamic World) | S1_log_VV_div_VH | 0.546 | 3.491 | 1.453 | 0.756 |
| Embeddings + region + Dynamic World | Ridge structural ratio (Embeddings + region + Dynamic World) | S1_VV+S1_VH | 0.546 | 3.491 | 1.453 | 0.756 |
| Embeddings + region + Dynamic World | Ridge direct ratio (Embeddings + region + Dynamic World) | S1_VV_div_VH_linear | 0.534 | 3.534 | 1.779 | 0.731 |

These baselines evaluate the physical linear VV/VH ratio reconstructed from the stored VV and VH columns. The dataset's original S1_VV_div_VH target remains the dB-space difference used in the main LightGBM tables above.

# Repeated Grouped-CV Stability

| target | feature_set | r2_mean | r2_std | rmse_mean | rmse_std | mae_mean | mae_std |
|---|---|---|---|---|---|---|---|
| S1_VH | Embeddings only | 0.939 | 0.008 | 1.223 | 0.089 | 0.921 | 0.062 |
| S1_VH | Embeddings + region + Dynamic World | 0.938 | 0.006 | 1.229 | 0.069 | 0.925 | 0.047 |
| S1_VV | Embeddings only | 0.910 | 0.008 | 1.145 | 0.042 | 0.832 | 0.043 |
| S1_VV | Embeddings + region + Dynamic World | 0.907 | 0.006 | 1.163 | 0.010 | 0.843 | 0.030 |
| S1_VV_div_VH | Embeddings only | 0.729 | 0.029 | 1.060 | 0.021 | 0.787 | 0.020 |
| S1_VV_div_VH | Embeddings + region + Dynamic World | 0.730 | 0.027 | 1.059 | 0.028 | 0.785 | 0.021 |

Each row summarizes 12 grouped validation folds: 4 folds across 3 random seeds.

# Regional S1_VV Diagnostics

| region | feature_set | r2 | rmse | mae | pearson_r | n_test |
|---|---|---|---|---|---|---|
| amazon_forest | Embeddings only | 0.933 | 1.152 | 0.785 | 0.966 | 216 |
| california_coast | Embeddings only | 0.920 | 1.041 | 0.756 | 0.960 | 216 |
| iowa_ag | Embeddings only | 0.850 | 1.308 | 0.758 | 0.929 | 216 |
| sf_bay_urban | Embeddings only | 0.868 | 1.533 | 1.037 | 0.934 | 216 |
| amazon_forest | Embeddings + region + Dynamic World | 0.935 | 1.129 | 0.778 | 0.967 | 216 |
| california_coast | Embeddings + region + Dynamic World | 0.922 | 1.029 | 0.750 | 0.961 | 216 |
| iowa_ag | Embeddings + region + Dynamic World | 0.839 | 1.357 | 0.789 | 0.923 | 216 |
| sf_bay_urban | Embeddings + region + Dynamic World | 0.878 | 1.474 | 0.981 | 0.940 | 216 |

Region-level diagnostics on the held-out test set.

# Land-use S1_VV Diagnostics

| dw_label_name | feature_set | r2 | rmse | mae | pearson_r | n_test |
|---|---|---|---|---|---|---|
| bare | Embeddings only | 0.665 | 1.254 | 0.863 | 0.831 | 96 |
| built | Embeddings only | 0.734 | 2.098 | 1.226 | 0.871 | 96 |
| crops | Embeddings only | 0.770 | 1.005 | 0.777 | 0.878 | 96 |
| flooded_vegetation | Embeddings only | 0.924 | 0.777 | 0.545 | 0.962 | 96 |
| grass | Embeddings only | 0.857 | 0.980 | 0.716 | 0.928 | 96 |
| shrub_and_scrub | Embeddings only | 0.784 | 1.357 | 0.888 | 0.891 | 96 |
| snow_and_ice | Embeddings only | 0.891 | 1.047 | 0.767 | 0.945 | 96 |
| trees | Embeddings only | 0.859 | 0.901 | 0.724 | 0.928 | 96 |
| water | Embeddings only | 0.841 | 1.504 | 1.002 | 0.923 | 96 |
| bare | Embeddings + region + Dynamic World | 0.699 | 1.189 | 0.826 | 0.849 | 96 |
| built | Embeddings + region + Dynamic World | 0.732 | 2.107 | 1.233 | 0.872 | 96 |
| crops | Embeddings + region + Dynamic World | 0.782 | 0.978 | 0.724 | 0.885 | 96 |
| flooded_vegetation | Embeddings + region + Dynamic World | 0.917 | 0.810 | 0.559 | 0.958 | 96 |
| grass | Embeddings + region + Dynamic World | 0.858 | 0.978 | 0.691 | 0.929 | 96 |
| shrub_and_scrub | Embeddings + region + Dynamic World | 0.788 | 1.345 | 0.894 | 0.893 | 96 |
| snow_and_ice | Embeddings + region + Dynamic World | 0.898 | 1.016 | 0.761 | 0.948 | 96 |
| trees | Embeddings + region + Dynamic World | 0.847 | 0.938 | 0.744 | 0.923 | 96 |
| water | Embeddings + region + Dynamic World | 0.850 | 1.462 | 0.988 | 0.931 | 96 |

Land-use diagnostics on the held-out test set.

# Feature Importance Summary

Top 5 features by permutation importance

Embeddings only

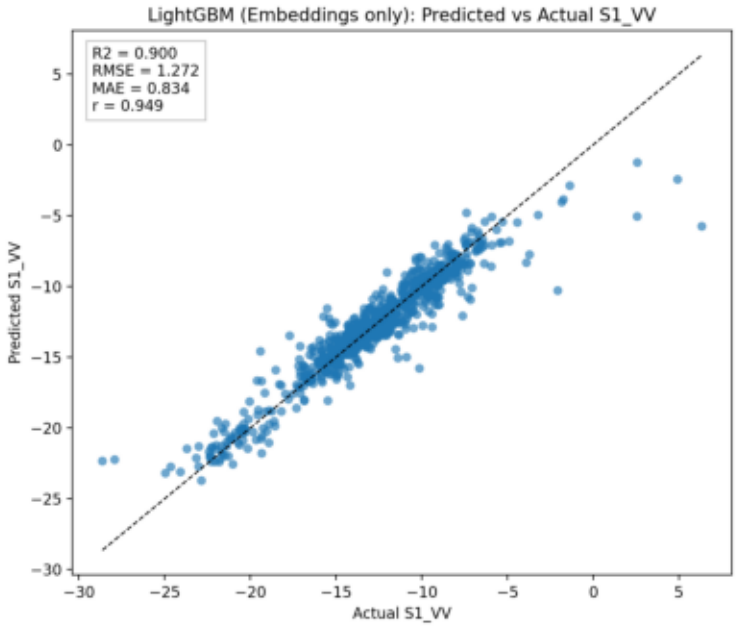
```
S1_VV: A25 (0.126), A27 (0.109), A49 (0.066), A17 (0.039), A19 (0.037)
S1_VH: A25 (0.268), A27 (0.066), A49 (0.030), A19 (0.028), A34 (0.023)
S1_VV_div_VH: A25 (0.126), A24 (0.047), A63 (0.042), A39 (0.035), A07 (0.033)
```

Embeddings + region + Dynamic World

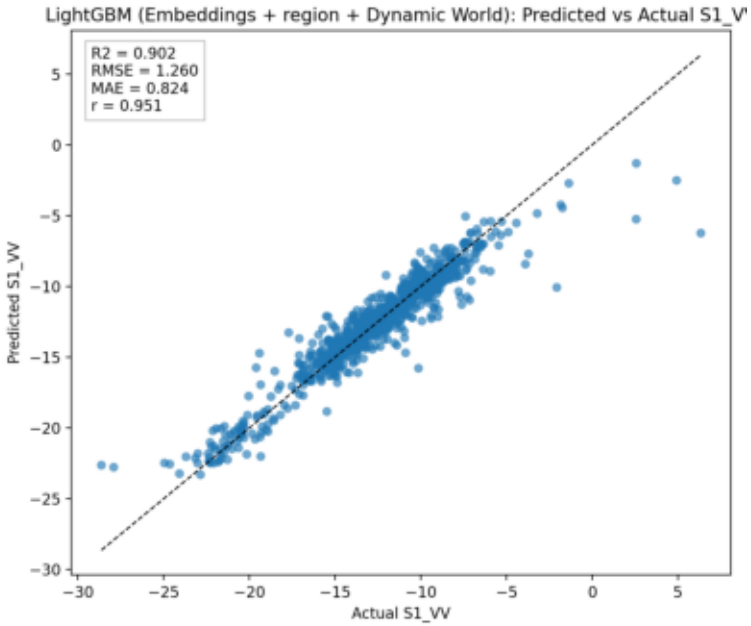
```
S1_VV: water (0.113), A27 (0.084), A49 (0.055), A25 (0.041), A17 (0.037)
S1_VH: water (0.117), A25 (0.105), A27 (0.045), bare (0.030), A49 (0.026)
S1_VV_div_VH: A25 (0.077), A31 (0.060), A39 (0.048), A63 (0.035), trees (0.028)
```

# Predicted vs Actual

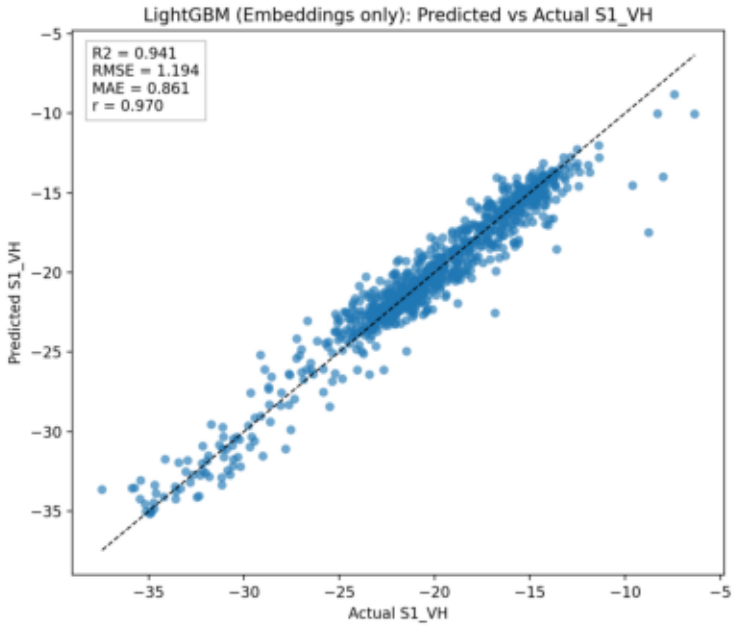
predicted vs actual embedding only S1 VV



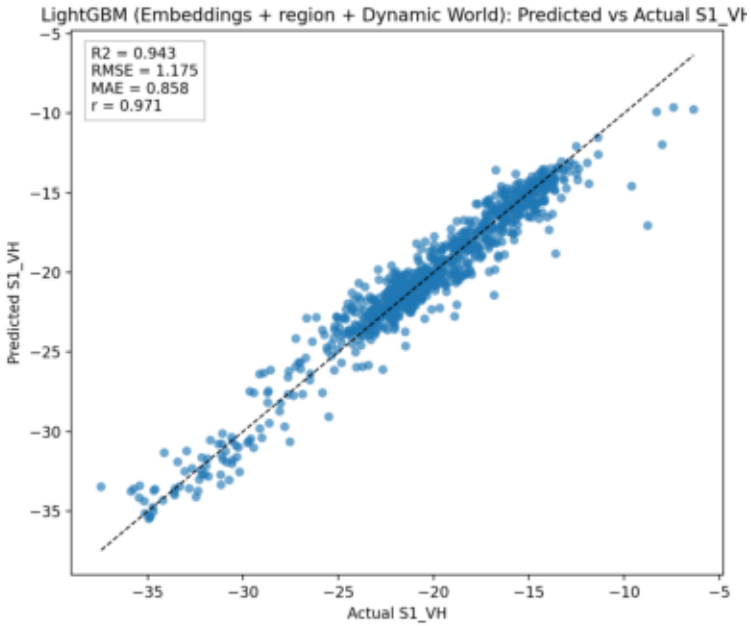

predicted vs actual embedding plus context S1 VV




predicted vs actual embedding only S1 VH

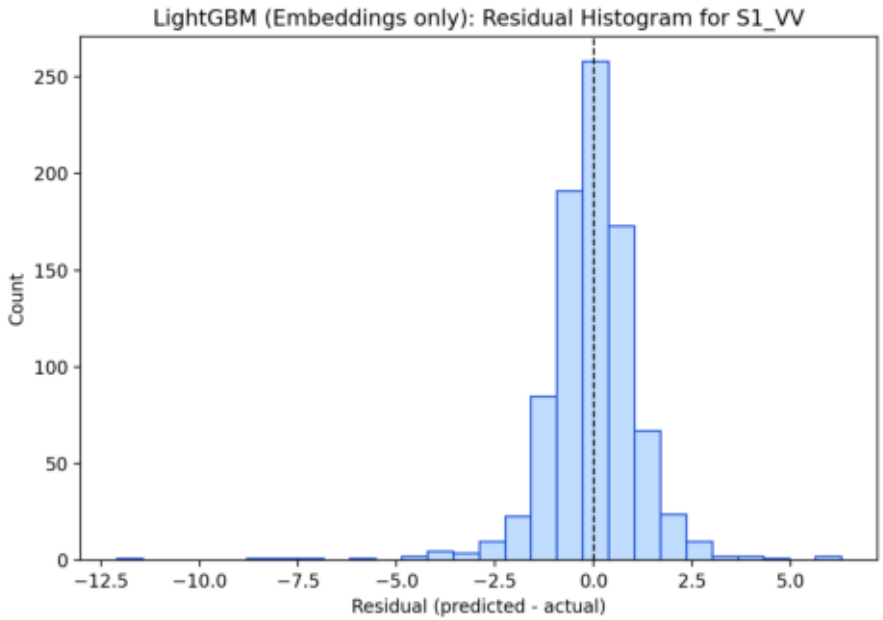



predicted vs actual embedding plus context S1 VH

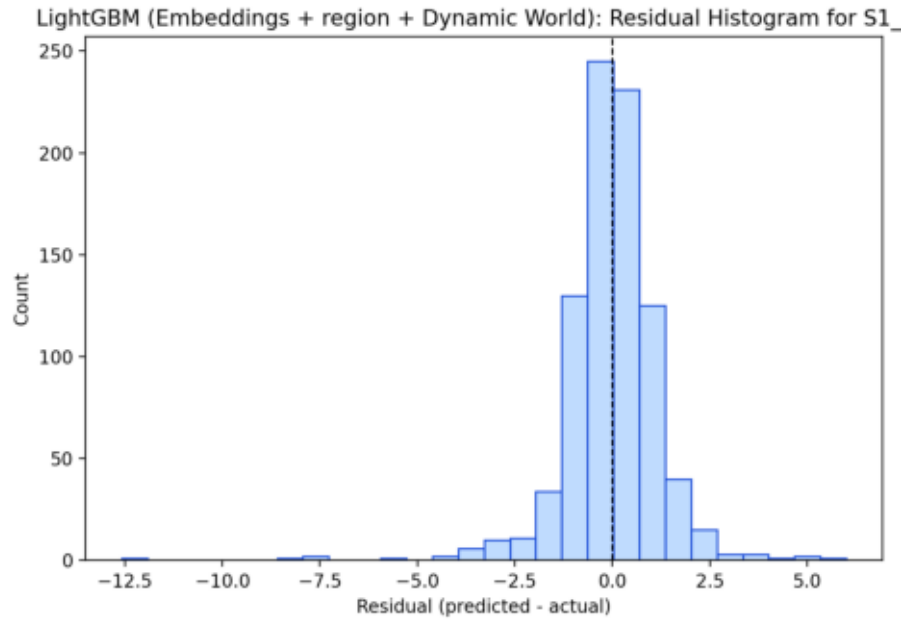


# Residual Histograms

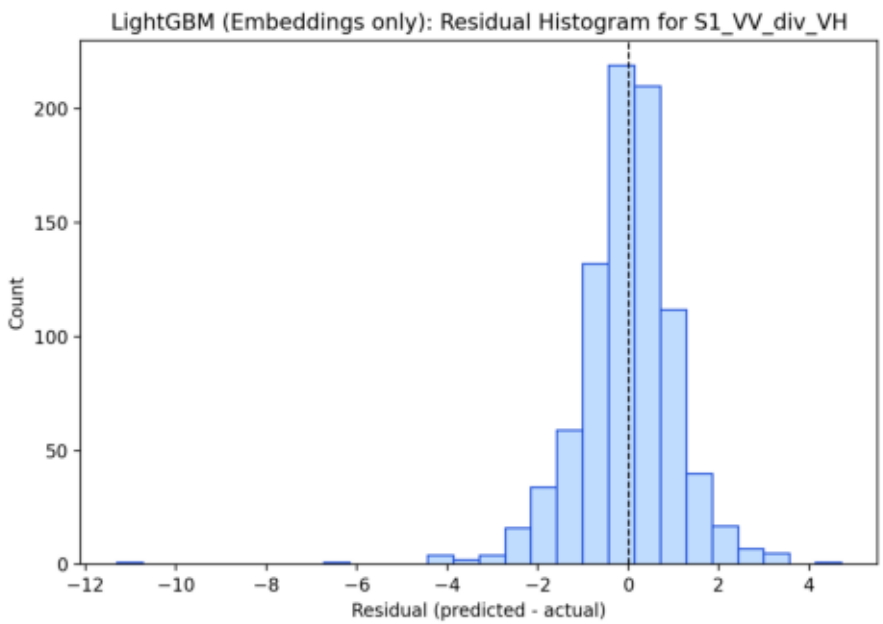
residual histogram embedding only S1 VV

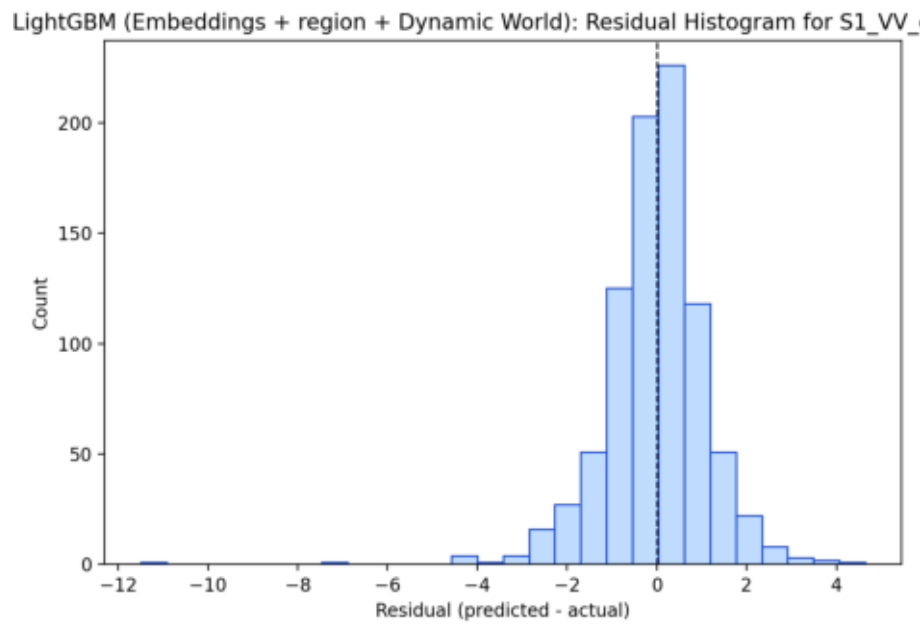



residual histogram embedding plus context S1 VV




residual histogram embedding only S1 VV div VH




residual histogram embedding plus context S1 VV div VH